Year 8 Science – Booklet 27: Chemistry

Chemical Reactions (3.1 Chemical Reactions) — ANSWERS

3.1 Chemical reactions – In-text questions

A. The atoms are rearranged to create new substances – they are joined together in one way before the reaction and in a different way after it (no atoms are created or destroyed, just rearranged).

B. Any three of: seeing flames or sparks; noticing a smell (sweet or foul); feeling the chemicals get hotter or colder; hearing a bang or gentle fizzing; seeing bubbles of gas; a colour change.

C. Any three of: medicines (e.g. paracetamol), fabrics (e.g. polyester), building materials (e.g. cement).

D. Any two of: melting, dissolving, freezing, boiling, evaporating.

Summary Questions

1. Choose the correct bold word (6 marks)

Chemical reactions involve re-arranging **atoms**. Chemical reactions **always** make new substances. They **are not** easily reversible. They **always** involve energy transfers. Physical changes include changes of **state**. They **are** reversible.

2. Chemical or physical change? (4 marks)

a) Burning diesel to make carbon dioxide and water – **Chemical**
b) Dissolving sugar in tea to make it taste sweet – **Physical**
c) Boiling water to make steam – **Physical**
d) Baking raw cake to make cooked cake – **Chemical**

3. Compare chemical changes with physical changes, with examples (6 marks QWC)

Chemical reactions create new substances by rearranging atoms and joining them together differently – e.g. burning wood produces ash, carbon dioxide and water, substances very different from the original wood, and this is not easily reversed. Physical changes, such as melting or dissolving, do not create new substances – the same substance is present before and after, just in a different state or form – e.g. melting ice to form water is easily reversed by freezing it again. Both types of change can involve energy transfers, but only chemical reactions make genuinely new substances and are difficult to reverse.

C3.1 Chemical reactions (workbook)

A. Fill in the gaps

A chemical reaction is a change that makes **new** substances. In a chemical reaction, the atoms in the starting substances are **rearranged** and join together **differently**. It is **not** easy to reverse a chemical reaction. Chemical reactions involve **energy** transfers to or from the surroundings. Chemists use **catalysts** to speed up reactions. Not all changes involve chemical reactions. A **physical** change, such as melting, is usually reversible.

B. Diagrams W, X, Y, Z (before/after a change)

a). **W and Z** show chemical reactions.

b). In W and Z, the atoms are rearranged and joined together in new combinations after the change – new bonds form between atoms that weren't grouped that way before. In X and Y, the particles keep the same groupings/bonds before and after (just repositioned or in a different state), showing these are physical changes rather than chemical reactions.

C. True for chemical reactions, physical changes, or both?

New substances are made – **chemical reactions only**
Easily reversible – **physical changes only**
Involves energy transfers – **both**
Atoms are rearranged and join together differently – **chemical reactions only**

D. a) Paragraph comparing physical changes with chemical reactions

Chemical reactions produce new substances by rearranging and rejoining atoms in different combinations, and are generally not easily reversed. Physical changes (melting, freezing, dissolving) do not produce new substances – the same substance simply changes state or form – and are generally easily reversible. Both types of change can transfer energy to or from the surroundings.

D. b) Explain the differences using ideas about atoms

In a chemical reaction, bonds between atoms break and new bonds form, joining the atoms in a different arrangement to make a new substance with different properties. In a physical change, the atoms (or molecules) stay bonded together in the same way – only their spacing, arrangement, or state changes – so the substance's identity stays the same, and reversing the change simply returns the particles to how they were, without needing to reform any chemical bonds.

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